



Explosion-Proof Track-hanging Robot - G7 Power



■ Product Overview

The robot adopts an explosion-proof modular high-power drive system that enhances safety, reliability, flexibility, and adaptability. The robot operates stably in temperature ranging from -20°C to 60°C, achieving ultra-long endurance of up to 10 kilometers and steep-slope climbing capability of 45°, with centimeter-level braking precision, making it suitable for complex track environments such as pipe galleries and tunnels. Equipped with a highly modular gimbal design, it can be flexibly configured for different inspection requirements. Its modular body design also supports rapid expansion with multiple sensor combinations, enabling versatile, multi-functional deployment across a wide range of intelligent inspection scenarios.

■ Features and Capabilities

Dimensions	≤1290 mm×380 mm×390 mm (L×W×H, subject to actual product)	Straight-Line Speed	0-2 m/s
Weight	≤50 kg (subject to actual product)	Turning Diameter	80 cm
Explosion-Proof Rating	Ex IIC T6 Gb/Ex IIIC T80°C Db	Climbing Ability	≤45°
Protection Rating	IP66	Charging Time	≤3 h
Drive Method	Dual Rail Drive	Endurance Capability	≤10 km
Braking Distance	Straight: ≤0.1 m, climbing: ≤0.1 m	Primary Navigation Method	Odometer + Hall Sensor + RFID
		Positioning Accuracy	≤±1 cm

■ Standard Features

Autonomous Inspection	Video Monitoring	Auxiliary Lighting
Autonomous Obstacle Avoidance	Offline Inspection	Status Indication
Remote Control	Audio-Visual Alarm	Autonomous Charging
Voice Intercom	Voice Broadcasting	One-Key Return

■ Optional Functions

Voiceprint Detection	Acoustic Imaging	Track Cleaning
Noise Detection	Laser Remote Measurement	Lifting Gimbal
Flame Detection	Body Cleaning	Optical Imaging

■ Scenario Algorithms

Personnel Detection Algorithm	Ammonia Synthesis Algorithm	Coke Oven Chamber Algorithm
Anomaly Detection Algorithm	Storage Tank Area Algorithm	Coke Oven Algorithm
Sulfurization Trench Algorithm	Compressor Plant Algorithm	Tank Car Algorithm
Evaporation Building Algorithm	Belt Conveyor Algorithm	Electrolytic Cell Algorithm
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